

. Cancer

Changes in Cancer Incidence Patterns Among a Northeastern American Indian Population: 1955-1969 Versus 1990-2004

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ABSTRACT: ***Purpose:** This manuscript examines shifts in patterns of cancer incidence among the Seneca Nation of Indians (SNI) for the interval 1955-1969 compared to 1990-2004. **Methods:** A retrospective cohort design was used to examine cancer incidence among the SNI during 2 time intervals: 1955-1969 and 1990-2004. Person-years at risk were multiplied by cancer incidence rates for New York State, exclusive of New York City, over 5-year intervals. A computer-aided match with the New York State Cancer Registry was used to identify incident cancers. Overall and site-specific standardized incidence ratios (SIRs = observed/expected $\times$ 100), and 95% confidence intervals (CIs), were calculated for both time periods. **Results:** During the earlier interval, deficits in overall cancer incidence were noted among males (SIR = 56, CI 36-82) and females (SIR = 71, CI 50-98), and for female breast cancers (SIR = 21, CI 4-62). During the more recent intervals, deficits in overall cancer incidence persisted among both genders (males SIR = 63, CI 52-77; females SIR = 67, CI 55-80). Deficits were also noted among males for cancers of the lung (SIR = 60, CI 33-98), prostate (SIR = 51, CI = 33-76) and bladder (SIR = 17, CI = 2-61) and among females for breast (SIR = 33, CI = 20-53) and uterus (SIR = 36, CI = 10-92). No cancer sites demonstrated increased incidence. Persons ages 60-69 years, 70-79 years, and ages 80+ years tended to exhibit deficits in overall incidence. **Conclusions:** Despite marked changes over time, deficits in overall cancer incidence have persisted between the time intervals studied. Tribal-specific cancer data are important for the development and implementation of comprehensive cancer control plans which align with local needs.*

have a large numbers of AIs and/or a population-based cancer registry.¹⁻⁹

While some commonalities exist among members of the 500-plus federally recognized tribal groups, each community embodies a unique set of experiences, environments and cultures resulting in distinct patterns of illness and deaths.¹⁰ Compared to the majority population, American Indians tend to be younger, less educated, have more poverty, and lower household incomes,¹¹ as well as a lower life expectancy.¹²

Despite a population of American Indians and Alaska Natives (ANs) which exceeds 4 million in the United States,¹³ a relatively limited number of publications have examined cancer patterns. Large databases maintained by the Indian Health Service (IHS) and the Surveillance, Epidemiology and End Results (SEER) program of the National Cancer Institute can be used to study incidence patterns. However, IHS data are limited to AIs who are eligible for and receive Indian Health benefits, while SEER is limited to selected geographic regions. In recent years IHS records have been linked to state-based cancer registries through the National Program of Cancer Registries to correct racial misclassification. In general, cancer patterns for specific AI groups are difficult to compare due to tribal heterogeneity, variable sizes of tribal groups, and variable time periods examined.

The present manuscript examines shifts in patterns of cancer incidence within the Seneca Nation of Indians (SNI) for the interval 1955-1969 compared to 1990-2004.

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Studies examining cancer incidence patterns among American Indians (AIs) have generally been limited to data for restricted time intervals and/or have been restricted to selected geographic regions which either

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Methods

Study Cohort. The cohort consisted of enrolled members of the Seneca Nation of Indians who were alive during any of the study years, which substantially expanded the Seneca cohort beyond that in an earlier study.⁴ The Seneca are members of the Iroquois Confederacy, which ranks as the 9th largest tribal group in the United States.¹¹

While the concept of “rural” defies description based on a single definition, features such as geographic location, population size, and density of inhabitants can be reviewed to characterize a community.¹⁴ The SNI maintain 2 reservation territories in southwestern New York State. These reservations are located about 50 miles (the Cattaraugus Reservation) and 70 miles (Allegany Reservation) from the nearest urbanized area in Buffalo, N.Y., and the majority of SNI members reside on or in proximity to these reservation areas. The current population (2000 Census) on each of these reservation areas is fewer than 2,500 persons, which is consistent with the US Census Bureau classification as rural. Population density is 69.9 persons per square mile on Cattaraugus and 30.2 persons per square mile on Allegany.

Study Design. We utilized a retrospective cohort design to examine patterns of cancer incidence during 1955-1969 and during 1990-2004. While cancer incidence among the SNI was assessed between 1955 and 2004, we focused on incidence during the periods 1955-1969 and 1990-2004 so as to contrast cancer patterns during two distinct time intervals. These intervals avoided the inclusion of calendar years during the 1970s and 1980s when the Seneca greatly expanded the availability of reservation-based medical centers. A dynamic cohort was used with individual cohort members added through births and removed at time of death. Membership was based on tribal roll book listings maintained by the Tribal Clerk’s Office. This research was approved by the SNI and also by our local institutional review board.

Each cohort member contributed 1 person-year to the appropriate sex and age category for each complete year of survival and 0.5 person-years during their year of birth or year of death. Endpoints for person-year calculations were: (1) date of death (or age 99 years, if date of death was unknown), or (2) survival thru December 31, 2004. Sex-specific person-year totals for the study cohort were accumulated over ten 5-year intervals (1955-1959, 1960-64, 1965-1969, . . . , and 1995-2004) and were partitioned into 5-year age categories (0-4 years, 5-9 years, . . . , 80-84 years, and 85-99 years). Cohort members who developed a

malignancy during the study period continued to contribute person-years at risk, allowing for the potential development of multiple primary malignancies.

Person-years at risk among the study cohort were multiplied by corresponding sex-specific, age-specific, site-specific and calendar year-specific cancer incidence rates for New York State, exclusive of New York City. Expected cancers were calculated during 5-year periods based on the mid-point of each interval (eg, calendar year 1957 for the interval 1955-1959, and 2002 for the interval 2000-2004), based on data from the New York State Cancer Registry (NYSCR) and Census enumerations of the state population, exclusive of New York City, during corresponding years. Expected cancer cases were summarized across each of the three 5-year time intervals of each 15-year period (eg, 1955-1969 = 1955-1959, 1960-1964 and 1965-1969).

Incident cancers among the SNI were identified through a computer match of cohort members with data files maintained by the NYSCR during each of the years included in the study. Cancer has been a reportable condition in New York State since 1940, when the NYSCR was established. Matching criteria included last name, first name (first 4 characters), gender, year of birth (± 5 years) and county of residence. All possible matches were manually checked for confirmation using other available data and allowed for accurate determination of true matches. During the study interval, cancer cases were classified at the NYSCR using various versions of the International Classifications of Diseases (ICD) (eg, ICD-7, ICD-8, ICD-9, ICD-10 and ICD-oncology, 2nd and 3rd editions). The NYSCR provided information on incident cancers diagnosed among cohort members using the topography, histology and behavior codes from the ICD for Oncology, Third Edition, US Interim Version 2000.¹⁵

Analyses. Standardized incidence ratios (SIRs) (based on the ratio of [observed cases/expected cases] multiplied by 100), were calculated for the intervals 1955-1969 and for 1990-2004, along with 95% confidence intervals (CIs) based on a Poisson distribution.¹⁶ A point estimate was considered to differ significantly from expected if its 95% CI did not include 100.

Results

The SNI cohort included a total of 54,473 person-years of observation during the 1955-1969 interval and 83,266 person-years of observation during the 1990-2004 interval. The shape of the population pyramids demonstrated a marked shift between time

periods, with increases in mid-adult and older adults during the later interval. The median ages among males increased from 23 to 29 years between the initial interval and the later period, while among females the median age changed from 24 to 33 years.

Among SNI males, the most commonly diagnosed cancers between 1955-1969 included colorectal cancers (n = 7, 27%) followed by oral cancers (n = 3, 12%) and cancers of the liver (n = 2; 8%), pancreas (n = 2, 8%), and prostate (n = 2, 8%), while during the interval 1990-2004 common malignancies included prostate cancers (n = 24, 23%), colorectal cancers (n = 29, 19%) and lung cancers (n = 15; 14%).

For females, the most commonly diagnosed cancers during the interval 1955-1969 included cervix (n = 8, 22%), colorectal (n = 6, 16%) and breast cancers (n = 3, 8%). During the interval 1990-2004, common malignancies among SNI females were lung cancers (n = 26, 22%), colorectal cancers (n = 19, 16%) and breast cancers (n = 18; 15%).

As shown in Table 1, males demonstrated an overall deficit of cancer cases for the period 1955-1969 (SIR = 56, CI 36-82). A deficit of lung cancer cases was also noted (SIR = 0, CI 0-44). No individual cancer sites were found to be significantly elevated during this time period. For the interval 1990-2004 males continued to exhibit a deficit of overall cancers (SIR = 63, CI 52-77). Site-specific deficits were noted for lung (SIR = 60, CI 33-98), prostate (SIR = 51, CI 33-76) and bladder cancers (SIR = 17, CI 2-61), while no site demonstrated a significant excess.

Females exhibited an overall deficit of cancer cases (SIR = 71, CI 50-98) as well as a deficit of breast cancer cases (SIR = 21, CI 4-62) during the period 1955-1969 (see Table 2). Between 1990 and 2004 females also demonstrated a deficit for malignancies overall (SIR = 67, CI 55-80) and site-specific deficits for cancers of the breast (SIR = 33, CI 20-53) and uterus (SIR = 36, CI 10-92). An excess of liver cancers was noted during this interval (SIR = 367, CI 100-940).

Table 3 presents cancer incidence patterns within age groups by gender and time period. During the earlier 15-year period, males demonstrated a deficit of cancer cases only for the age group 50-59 years (SIR = 0; CI 0-43), while during the latter 15-year interval, significant deficits in cancer incidence were noted for males ages 60-69 years (SIR = 64, CI 43-92), 70-79 years (SIR = 61; CI 40-90), and ages 80+ years (SIR = 41; CI 20-73). Between 1955 and 1969, cancer incidence deficits were observed among females ages 60-69 years (SIR = 24, CI 5-71) and ages 70-79 years (SIR = 28; CI 6-83), while from 1990-2004 incidence deficits were observed for age groups 70-79 years (SIR = 54; CI = 33-83) and ages 80 and above (SIR = 53; CI = 30-86). No significant excesses were observed for any of the age categories.

Discussion

The recently released Report to the Nation on the Status of Cancer provided an update on cancer among AIs and ANs.¹⁷ While comprehensive in scope, cancer incidence data was based on SEER data, as well as state

Table 1. Standardized Incidence Ratios for Malignancies Among Seneca Nation of Indians Males: 1955-1969 and 1990-2004

Cancer Site	Time Period: 1955-1969				Time Period: 1990-2004			
	Observed	Expected	SIR	95% CI	Observed	Expected	SIR	95% CI
All Sites	26	46.7	56	36-82	104	163.9	63	52-77
Oral	3	2.3	129	27-378	7	4.4	158	64-326
Colorectal	7	7.7	91	37-188	20	18.9	106	65-163
Liver	2	0.4	535	65-1,932	4	1.9	206	56-527
Gallbladder	0	0.4	0	0-906	1	0.3	389	10-2,168
Pancreas	2	1.6	123	15-445	5	3.8	132	43-309
Lung	0	8.3	0	0-44	15	25.2	60	33-98
Prostate	2	5.5	36	4-131	24	47.2	51	33-76
Bladder	0	3.1	0	0-117	2	11.9	17	2-61
Hodgkin & non-Hodgkin Lymphoma	1	3.3	30	1-168	7	8.5	82	33-169
Leukemia	0	2.1	0	0-177	6	5.1	117	43-255
Other	9				20			

Statistically significant results are highlighted.

Table 2. Standardized Incidence Ratios for Malignancies Among Seneca Nation of Indians Females: 1955-1969 and 1990-2004

Cancer Site	Time Period: 1955-1969				Time Period: 1990-2004			
	Observed	Expected	SIR	95% CI	Observed	Expected	SIR	95% CI
All Sites	37	52.2	71	50-98	118	176.9	67	55-80
Stomach	1	1.7	58	1-326	6	2.3	263	97-573
Colorectal	6	8.2	73	27-158	19	20.7	92	55-144
Liver	1	0.2	429	11-2,391	4	1.1	367	100-940
Gallbladder	0	0.7	0	0-544	1	0.6	159	4-886
Larynx	0	0.1	0	0-2,580	2	1.0	201	24-725
Lung	1	1.6	62	2-347	26	23.1	112	73-165
Breast	3	14.1	21	4-62	18	54.1	33	20-53
Cervix	8	4.0	201	87-397	4	3.5	115	31-295
Uterus	1	3.9	25	1-142	4	11.1	36	10-92
Ovary	1	2.9	34	1-192	4	6.8	59	16-152
Bladder	0	1.2	0	0-297	2	4.9	41	5-149
Hodgkin & non-Hodgkin Lymphoma	1	2.5	40	1-221	4	8.1	49	13-126
Leukemia	2	1.6	129	16-467	5	4.3	117	38-274
Other	12				21			

Statistically significant results are highlighted.

cancer registry records linked with IHS patient registration files. While this report represents a significant addition to our understanding of cancer occurrence in AI populations, it does have a number of limitations. IHS data are limited to enrolled members of federally recognized tribes whose treatment was underwritten by IHS (eg, excludes third-party insurance). Moreover, cancer incidence data in the

report are essentially cross-sectional during a limited time interval (1999-2004) and based upon a subset of all AIs. Although acknowledging regional variations in cancer patterns, this report¹⁷ also omits reference to specific northeastern tribal populations.

Results from the current study document overall deficits in cancer incidence among the SNI which have continued from an early interval (1955-1969) through a

Table 3. Standardized Incidence Ratios for Malignancies Among Seneca Nation of Indians, by Gender and Age Group: 1955-1969 and 1990-2004

Age Group	Time Period: 1955-1969				Time Period: 1990-2004			
	Males		Females		Males		Females	
	SIR	95% CI	SIR	95% CI	SIR	95% CI	SIR	95% CI
0-9 ys	273	33-985	157	4-873	94	2-523	89	2-495
10-19 ys	0	0-631	0	0-632	165	20-595	300	62-878
20-29 ys	166	20-599	0	0-254	0	0-139	33	1-186
30-39 ys	0	0-228	115	31-294	95	31-222	68	25-147
40-49 yrs	27	1-151	135	67-242	87	40-165	60	31-105
50-59 ys	0	0-43	101	51-181	73	45-113	68	43-102
60-69 ys	73	35-135	24	5-71	64	43-92	87	61-120
70-79 ys	59	24-122	28	6-83	61	40-90	54	33-83
80+ ys	81	22-208	98	27-250	41	20-73	53	30-86
Overall	56	36-82	71	50-98	63	52-77	67	55-80

Statistically significant results are highlighted.

more recent period (1990-2004). We are unaware of comparable data on cancer incidence patterns among AIs dating back to the earlier period. The cancer deficits during earlier years have generally been attributed to consumption of a native diet, natural living and premature deaths. Throughout the 1970s and 1980s the Seneca experienced considerable positive changes, including the emergence of reservation-based medical centers. Interestingly, data from national surveys noting greater rates of cancer risk factors among AIs would have suggested elevated cancer incidence.^{11,17,18}

A prior examination of cancer incidence patterns among the Seneca Nation from 1955-1984, noted significantly lower overall cancer incidence for both genders, as well as for cancers of the lung and bladder among males and among females for cancers of the colon, breast, and bladder.⁴ Deficits in lung cancer incidence were noted among males for both study intervals, along with a deficit of male bladder cancers between 1990 and 2004, which might suggest lower rates of smoking compared to the general population of New York State. In contrast, the incidence of lung cancer among Seneca females did not differ from expected. Deficits in lung cancer incidence have been noted among male but not among female AIs,^{5,8,9} although others have noted lung cancer deficits among both genders.^{6,19,20} Also, considerable regional variation in AI and AN lung cancer incidence patterns has been acknowledged.^{17,20,21} Data on health behaviors among the SNI are not well described. National survey data show that current smoking rates among AI/AN exceed that of non-Hispanic whites^{17,18} but may not be representative of the Seneca.

Disease patterns for cancer sites amenable to early detection did not follow any clear pattern. Colorectal cancer incidence among the SNI did not vary significantly by time period or gender. National surveys indicated that rates of colorectal cancer screening among AI/ANs lag behind those for non-Hispanic whites during more recent calendar years.^{17,18} Similar to the observation of breast cancer deficits in other AI groups^{5,6,8,17,19} deficits in breast cancer incidence were noted among the SNI for both time intervals. Breast cancer incidence rates vary by age group, race/ethnicity, and multiple other risk factors such as family history, reproductive factors (eg, late age at first live birth), breastfeeding history, menstrual history (eg, early age at menarche, late menopause), obesity, level of exercise, alcohol consumption, and use of exogenous hormones.^{22,23} While detailed information on fertility and reproductive patterns are not available, US Census data suggest that current fertility rates among AI/AN women are higher compared to non-Hispanic white women¹¹ which may partially

explain the lower rate of breast cancer. Contemporary rates of mammography among AI/AN women (69% within the last 2 years) tend to be slightly lower than among white women (76%).¹⁷

The SIR for prostate cancer increased from 36 (n = 2 cases) to 51 (n = 24 cases) between the 2 time periods. While this suggests an apparent difference, it was not large enough to be statistically significant. National data indicate that AI/AN males tend to complete prostate specific antigen testing less often compared to non-Hispanic white males (49% vs 58%, respectively).¹⁷ There is some evidence to suggest the occurrence of more advanced disease among Seneca during the more recent years. Unfortunately, staging data was not available for the earlier period. Decreased prostate cancer incidence has also been observed in other AI groups.^{5,6,8,17,19,24}

Females demonstrated an excess of liver cancer between 1990 and 2004. While hepatobiliary cancers have been generally increased among AI populations,^{3,5,6,8-10,17,25} it is difficult to assess the significance of this observation in the present study based on a limited number of cases. Multiple risk factors for liver cancer have been identified and the presence of these risk factors among these cases is unknown. Moreover, it is possible that some of these cases could have represented metastatic disease.

This study was unable to assess information on cancer risk behaviors, lifestyle exposures, environmental exposures, genetic factors and/or occupational exposures. While the robust size of the Seneca Nation facilitated tribal-specific analyses, interpretation of several of the point estimates for cancer incidence remains a challenge, especially across gender, time interval and site groupings. Since the NYSCR did not include cases among New York City residents before 1976, it is possible that incident cancer cases may have been under-reported prior to statewide case reporting. Unique strengths of this study were the use of tribal roll books to identify tribal members thus eliminating the issues of racial misclassification, the sizable number of person-years of observation accrued and matching the cohort against NYSCR records; also the NYSCR is a longstanding member of the National Program of Cancer Registries. The retrospective cohort design is particularly well suited to this application.

The deficits in cancer noted among the SNI in older age categories is noteworthy since the analytic approach adjusts for differences in the age distribution of the study population despite increasing numbers of persons reaching age 50 years and beyond during the 1990-2004 study interval. It remains possible that cohort members, who were at high risk of eventually developing cancer, could have succumbed to other

non-cancer causes of mortality (eg, injuries, diabetes, heart disease) prior to reaching ages at which they would have developed malignancies. Similar age-specific deficits in cancer incidence were noted among Manitoba Indians between 1970 and 1979.⁸ These issues may be more fully examined through studies of mortality currently under way.

Despite marked changes over time, deficits in overall cancer incidence have persisted between the time intervals studied. Of note, deficits were apparent for breast and prostate cancers between 1990 and 2004. Observational studies are important in documenting health disparities and aid in focusing resource allocation and/or policy changes to improve health outcomes. Approaches emphasizing both primary prevention (eg, smoking prevention/cessation, healthy body weight) and secondary prevention (eg, screening) are likely to generate significant changes in disease outcomes as well as improvements in quality of life. Tribal-specific data on cancer patterns are important for the development and implementation of comprehensive cancer control plans focused on primary, secondary and tertiary cancer prevention strategies which align with local needs.

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